

MIGUEL YU

Madrid, Madrid, Spain

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Eligible to sign internship agreement via IE University

SUMMARY

Dual-degree student (PPLE + Data & Business Analytics) focused on AI infrastructure, data-intensive systems, and performance-aware computing for scientific workflows. Experience in Python/Julia, distributed data processing, and benchmarking; interested in sustainable AI/HPC infrastructures and large-scale research systems).

RESEARCH INTERESTS

High-performance machine learning and hardware-aware optimisation; Distributed data systems for scientific computing; AI workflows in large-scale research infrastructures; ML applications in healthcare and public-sector systems.

- Target domains: sustainable AI infrastructures & techniques, HPC-enabled ML workflows, digital twin and data-intensive applications.

EDUCATION

IE UNIVERSITY (IEU)

Dual Bachelor in Philosophy, Politics, Law & Economics and Data & Business Analytics

Madrid, Spain

Expected: Jul 2028

- GPA: 8.03/10.00
- Awarded the IE Collaboration Scholarship.
- Relevant subjects: "Operating Systems & Parallel Computing" (concurrency, scheduling, scalability analysis, intro GPU/accelerators), "Big Data Technologies" (SQL/NoSQL, Spark/PySpark, storage architectures), "AI/ML Foundations" & "... Analytics" (model evaluation, deep learning fundamentals, transformers overview)

SEK-CATALUNYA INTERNATIONAL SCHOOL

International Baccalaureate (IB) Diploma Program

Barcelona, Spain

Graduated: Jun 2023

- Bilingual diploma: English and Spanish.

EXPERIENCE

EXTERN

Remote

Healthcare Data Analytics Externship — TruBridge

Dec 2025 – Present

- Analysing healthcare datasets to extract operational and clinical insights using Python and SQL, with emphasis on data quality, and inference.
- Applying statistical and ML techniques to support data-driven rural healthcare systems decision-making.

IEU POLICY LAB

Segovia, Spain

Writer and Researcher

Jan 2024 - Jun 2024

- Researched technology policy options and drafted a technical consultancy report for the IAEA Technical Cooperation Department, assessing existing and emerging technologies and implementation constraints.

RESEARCH & TECHNICAL PROJECTS

Hardware-Aware Optimisation of Machine Learning Training in Julia

Personal Project

Julia, Flux.jl, Parallel Computing

Jan 2025 – Present

- Implementing and benchmarking ML training workflows in Julia (Flux), analysing performance trade-offs across batch sizes, memory allocation strategies, and hardware utilisation.
- Conducting reproducible performance benchmarking to compare CPU single-threaded and multi-threaded performance.

AI/ML - Healthcare Retrieval-Augmented Generation System

Course Project

Python, React, HuggingFace

Sep 2025 – Dec 2025

- Built a modular RAG pipeline for healthcare documents (PDF + tabular databases): ingestion, chunking, embeddings, retrieval, and LLM response generation.
- Evaluated retrieval quality and end-to-end response relevance; identified scaling considerations (index size, latency, caching, embedding optimisation).

EXTRACURRICULAR ACTIVITIES

- **IE ManagementxTechnology (MxT) Program:** Accelerating skills and mindset to take advantage of rapid advancements and disruptive technologies to “innovate, launch and lead”, within a new generation of entrepreneurship. (Jan 2025-)
- **Governance of Change Club — Research Officer:** Researching on technology governance and institutional change. (Oct 2025-)
- **Class Representative:** Elected by and to represent a class of 8 students, in communication with Programme Management and Student Government. (Oct 2023)

LANGUAGES

- **Chinese** (Native). **Spanish** (Native). **English** (Fluent, Cambridge C1 & IB Eng. A L&L HL). **Catalan** (Advanced, C1 from Catalan territory high school graduation).

ADDITIONAL INFORMATION

- **Computer Skills:**
 - **Languages:** Python, Julia, SQL (MySQL, NoSQL), R
 - **Data / Distributed:** Spark (PySpark), Hadoop/MapReduce, MongoDB, data warehousing concepts
 - **ML:** scikit-learn; TensorFlow/Keras or PyTorch; Hugging Face
 - **Systems / HPC Concepts:** Linux/UNIX CLI, concurrency, scheduling, profiling/benchmarking, parallel patterns (threads/processes), cloud compute basics
 - **Tools:** Git/GitHub, Jupyter, PyCharm
- **Other Skills:** Technical communication: policy consultancy report (IAEA) and project presentations. Open-minded; Communication, Leadership & Teamwork (Healthcare RAG); Conflict Resolution (Class Representative, and Other courses).
- **Interests:** Reading and drawing.